

AI Speaking Assistant

Use-Case Specification: UC5.2 Generate Appropriate Reply

Version: 1.0

Revision History

Date	Version	Description	Author
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1. Use-Case Name

UC5.2: Generate appropriate reply

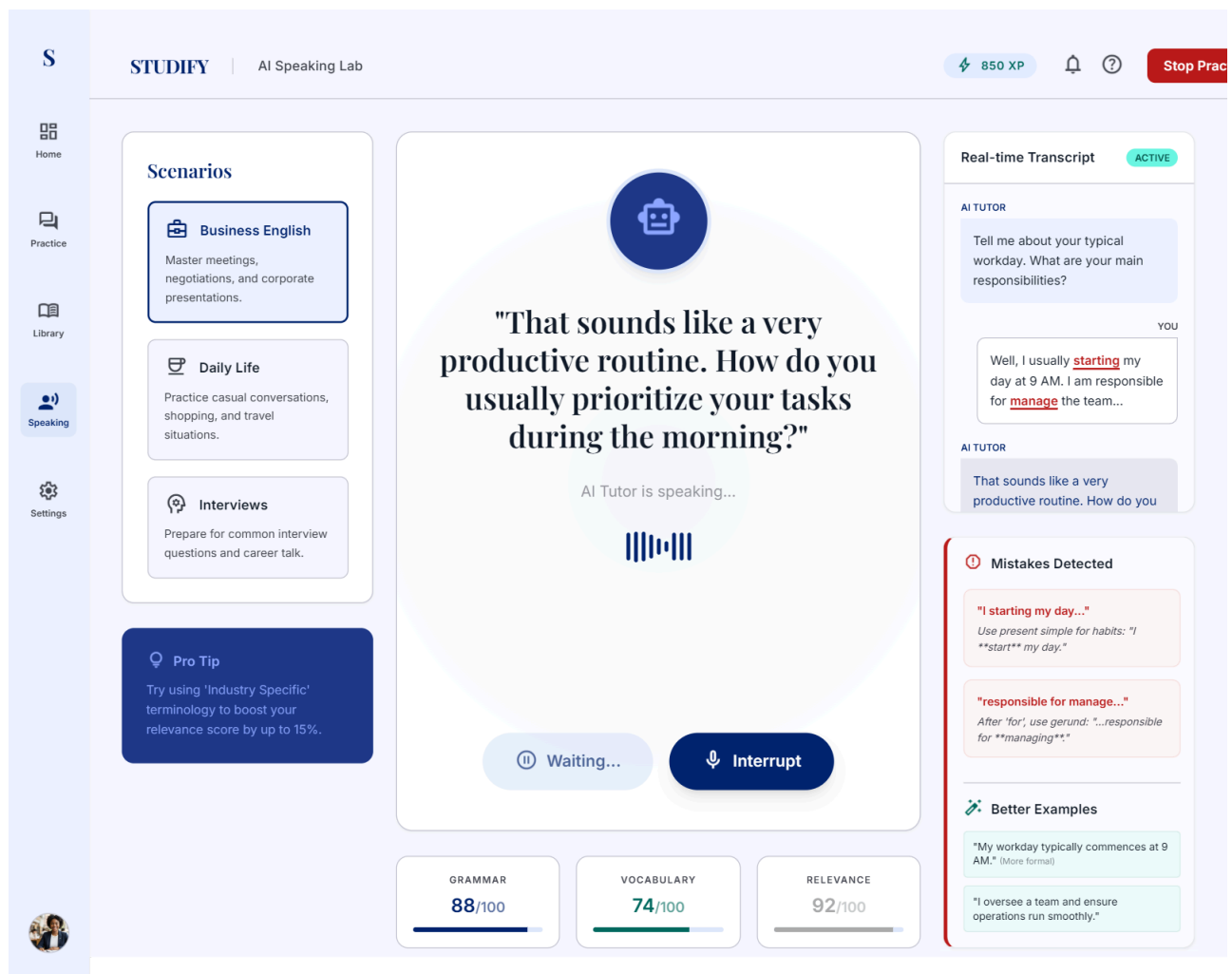
1.1 Brief Description

This use case captures the end-to-end process in which the AI Speaking Assistant receives the Learner's transcribed input, evaluates the context of the active practice scenario (e.g., Business, Casual, Interview), and uses the AI Engine to construct a natural, contextually relevant response in both text and synthesized audio forms.

2. Flow of Events

2.1 Basic Flow

1. The Learner speaks their turn into the interface.
2. The system executes **UC5.1: Speech recognition** to obtain the transcribed text transcript (<<include>>).
3. The system bundles the transcript, active scenario metadata, and prior conversation turns, submitting the payload to the AI Engine.
4. The AI Engine processes the request and generates an in-context conversational reply.
5. The system synthesizes the text response into natural-sounding voice audio.
6. The system displays the conversation text in the chat panel and plays the audio reply to the Learner.



2.2 Alternative Flows

2.2.1 Off-Topic Input Handling

1. In step 4 of the Basic Flow, the AI Engine detects that the user's input strays significantly from the scenario prompt.
2. The AI Engine generates a polite redirection response acknowledging the input while prompting the Learner back to the scenario core topic.
3. The execution resumes at step 5 of the Basic Flow.

The screenshot displays the STUDIFY AI Speaking Tutor interface. The top navigation bar includes the STUDIFY logo, the title "AI Speaking Tutor", and a "Stop Practice" button. A left sidebar contains icons for Dashboard, Practice, Library, Speaking, and Stats. The main content area is divided into three sections:

- Scenarios:** A list of three scenarios: "Business English" (Master meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel), and "Interviews" (Prepare for common interview questions and career talk).
- Real-time Transcript:** A section showing the user's input and the AI tutor's response. The user's input is "Tell me about your typical workday. What are your main responsibilities?". The AI tutor's response is "Well, I usually **starting** my day at 9 AM. I am responsible for manage the team...". Below this, the AI tutor says "That sounds interesting. How do you prioritize your tasks?".
- Live Corrections:** A section showing two errors: "Grammar Error" ("I starting my day...") and "Prepositional Error" ("...responsible for manage...").

The central area features a large blue circle and the text: "I appreciate the thought, but let's try to focus on describing your professional role. What are your main responsibilities?". Below this, it says "I'm listening... Speak clearly for the best transcription." and a "Listening..." button with a microphone icon.

At the bottom, there are three performance metrics: GRAMMAR 88/100, VOCABULARY 74/100, and RELEVANCE 92/100. A "SOPHISTICATED ALTERNATIVES" section is also visible at the bottom right.

2.2.2 AI Service Timeout

1. In step 4 of the Basic Flow, the connection to the AI Engine service times out (exceeds 5 seconds).
2. The system presents an error alert: "Response generation timed out. Click to retry."

3. The Learner clicks retry, and execution restarts at step 3 of the Basic Flow.

The screenshot displays the STUDIFY Practice Session interface. A central modal window shows a red circular icon with a white 'X' and the text: "Response generation timed out. We couldn't connect to our AI tutors. Please check your connection and try again." Below this is a blue "Retry" button and the error code "ERR_AI_TIMEOUT_02".

On the left sidebar, there are icons for Scenarios, Real-time Transcript, Mistakes Detected, and Better Examples. The main content area on the left lists three scenarios: "Business English" (Master meetings, negotiations, and corporate presentations), "Daily Life" (Practice casual conversations, shopping, and travel situations), and "Interviews" (Prepare for common interview questions and career talk). Below these is a "PRO TIP" box: "Try using 'Industry Specific' terminology to boost your relevance score by up to 15%."

On the right, the "REAL-TIME TRANSCRIPT" section shows an "AI TUTOR" prompt: "Tell me about your typical workday. What are your main responsibilities?" and a "YOU" response: "Well, I usually **starting** my day at 9 AM. I am responsible for manage the team..." Below this, the "MISTAKES DETECTED" section highlights two errors: "I starting my day..." (Use present simple for habits: "I **start** my day.") and "responsible for manage..." (After 'for', use gerund: "...responsible for **managing**").

At the bottom, a progress bar shows scores for Grammar (88/100), Vocabulary (74/100), and Relevance (92/100).

2.2.3 Inappropriate or Flagged Content

1. In step 4 of the Basic Flow, safety filters flag user input containing abusive, hateful, or explicit language.
2. The system bypasses normal conversation generation and responds with a standard safety prompt: *"Let's keep our conversation respectful and focused on the practice topic."*

3. The execution resumes at step 5 of the Basic Flow.

STUDIFY AI | Speaking Practice / Session 22

Scenarios

- Business English**
Master meetings, negotiations, and corporate presentations.
- Daily Life**
Practice casual conversations and travel situations.
- Interviews**
Prepare for common interview questions and career talk.

Real-time Transcript ACTIVE

AI TUTOR
Tell me about your typical workday. What are your main responsibilities?

YOU
Well, I usually **starting** my day at 9 AM. I am responsible for manage the team.

Mistakes Detected
Feedback is unavailable for flagged segments.

Reference Phrases

ALTERNATIVE 1
"My workday typically commences at 9 AM."
(More formal)

ALTERNATIVE 2
"I oversee a team and ensure operations run smoothly."

Pro Tip
Try using 'Industry Specific' terminology to boost your relevance score by up to 15%.

GRAMMAR
88 /100

VOCABULARY
74 /100

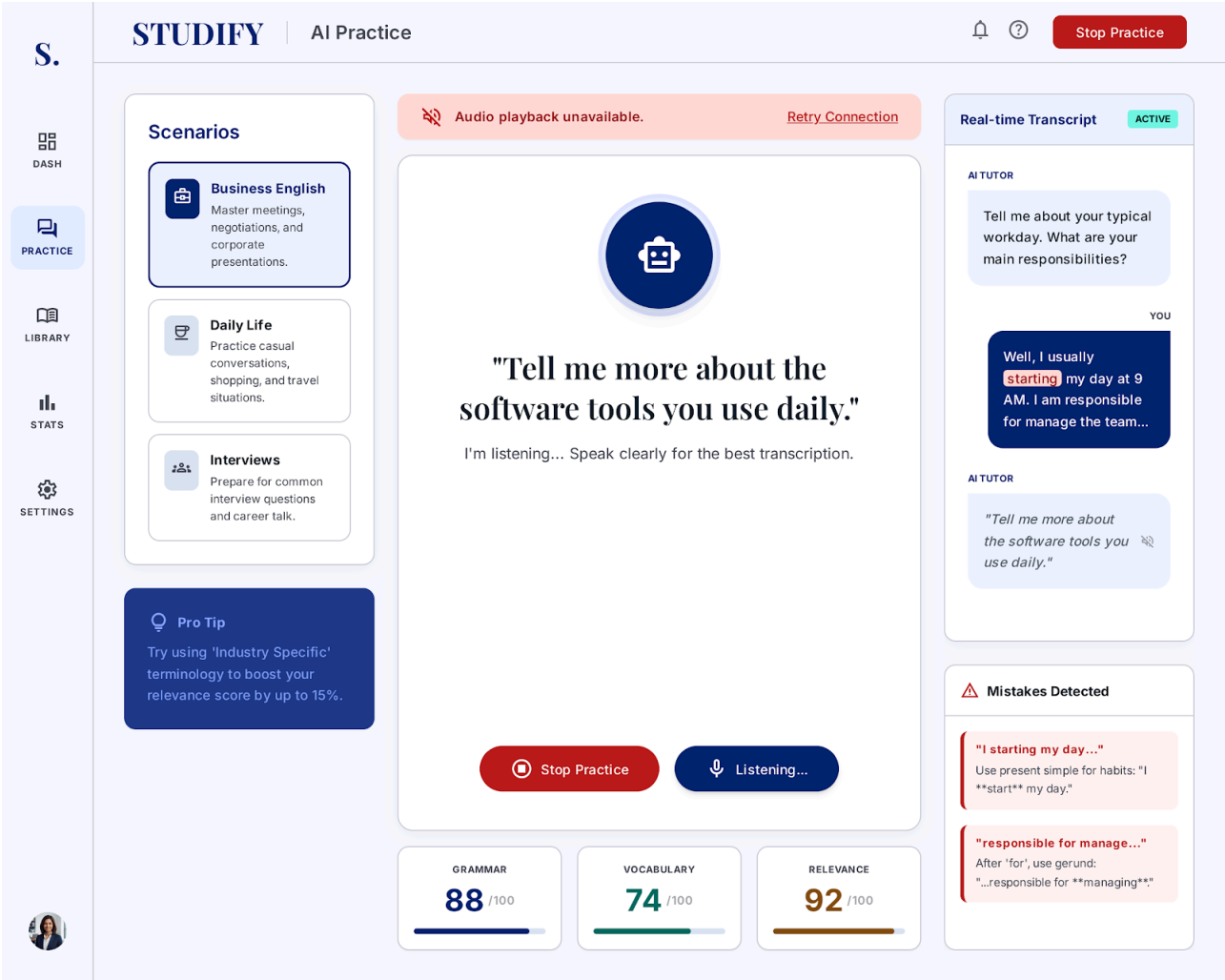
RELEVANCE
92 /100

Try Another Response

2.2.4 Text-to-Speech (TTS) Synthesis Failure

1. In step 5 of the Basic Flow, the TTS engine fails to synthesize audio.
2. The system displays the text reply in the chat interface and displays a non-intrusive notification:
"Audio playback unavailable."

3. The flow completes without audio playback.



3. Special Requirements

3.1 Conversational Latency

The total latency for reply text generation and Text-To-Speech (TTS) rendering combined must not exceed 3 to 5 seconds to maintain natural conversation dynamics.

3.2 Natural Voice Synthesis Quality

Audio output must utilize neural TTS voices sampled at a minimum of 24kHz to ensure realistic human cadence, tone, and inflection.

3.3 Conversation Context Length

The AI Engine must maintain and process a context history window of at least 10 prior dialogue turns to ensure coherent multi-turn conversations.

4. Preconditions

4.1 Active Scenario Context

A practice scenario must be initialized, and the dialogue state must be active.

4.2 Transcribed Text Input Received

Valid text output from **UC5.1** must be passed into the dialogue handler.

4.3 AI Engine Service Availability

The system's backend AI model API must be online and responsive.

5. Postconditions

5.1 Dialogue State Updated

The newly generated AI turn (text and audio) is saved into the session history, and the state advances to wait for the user's next turn.

5.2 Audio Response Cached

Synthesized voice audio files are temporarily cached on the client to allow instant replay by the user.

5.3 History Stack Updated

The full transcript exchange is appended to the user's session history record.

6. Extension Points

6.1 None

This use case contains no extension points.